Overview of the buffers in the PLC

Every message that can possibly be sent containing DataWords (DW) will ALWAYS include the index the data does not need to be consecutive, unless required. (this does not apply to the other types such as CW or SW's which are not indexed and fixed and are transacted as a whole.

The following buffers are for example how we organize memory in our plc's M2M solution but don't necessarily have to be identical, except that the protocol messaging content obviously needs to match. FuncW for example is not ever transmitted on the protocol, but having something like this in the sensor would help matters, but perhaps the sensor may only need one type fit for all etc.

Data Map

Converts physical PLC words to logical RTU words.

Physical PLC

	FuncW	Index	Unsol Mask	Hist Mask	
0	0C00	03E8	00FF	0F00	D1000 *Dig16*
1	6C00	03A9			
2		000F			D0016 *Dig16*
n					

Instantaneous (INTERROGATE) Buffer:

Logical RTU

	xxxx	
1	0000	DW1000
2	0000	DW937
3	0000	DW16
n		

Addressing notes:
Starting index index starts anywhere the protocol ensures index is always transferred with the value.

DWORDs are addressed in adroit by DDx
WORDs are addressed in adroit by DWx

Addressing
DWx
DDx
Dix
DLx
CWx
Awx

FuncW:

0 : TYPE *Msb {00(Dig),01(Int),10(ana)}*
1 : TYPE *Lsb*
2 : MEMTYPE *Msb {00(Dx),,10(DDx)}*
3 : MEMTYPE *Lsb*
4 : En Unsolicited reporting
5 : En History reporting
6 – 15 : null

Index:

Offset 0 - 65535

Unsolicited Mask:

Bitwise mask bits 0-15
Register types >0

History Mask:

Bitwise mask bits 0-15
Register types >0

Communications Buffers:
Priority:
History:

Head Idx	Tail Idx	Size (words)	0000

Fixed buffer control header
Not sent to master aswell.

TimeObj:
Msw +Lsw

TimeObj Msw	TimeObj Lsw	Index	Value	
XXXX	XXXX	0	0001	Msw
0000	0000	0	0002	Lsw
XXXX	XXXY	1	1234	
XXXX	XXXZ	2	FFFF	

DWORD/
LONG
Special case

QUEUE based buffer

Emptied from the oldest to the newset.

New records are pushed in at the back , transact data is acquired from the front.

Sent in fixed records where a record ios a whole row (4Words) or (8Words) if a 32 item is sent.
DWORD never get wrapped halvies… enough space is poped for oth to be ordered.
DWORDS are never transmitted halvies.

NOTES:
If during popping to wrap we pop the msw of a double the lsw wont be removed also the driver will detect the null time and null index!

Application Protocol:

Binary
All data is represented in Hexadecimal.
TCP connection established from the slave.

FunctionWord: *(used with all transactions)*

000x --- > FUNCTION …. { Action requires such as interrogate }
00x0 --- > DIRECTION (0=initiated from master, 1=initiated from slave(unsolicited, or link so far))
01xx <---- ACK …. acknowledge reception.
10xx <---- NACK .... nacknowledge reception.

Error Word: *(used with all nak transactions master and slave)*

0001 --- > Invalid Function
0002 --- > Invalid Data
0003 --- > Not enabled
0004 --- > Need time first.
0005 --- > No data available {perhaps a nak if say hist buffer has no events}
0006 --- > Huh?

ActionWord 0: *{sent from master}*

0 : START RTU
1 : RESET RTU {clear all buffers etc}
2 : GO LIVE { push inst buffer periodically}
3 : ENABLE-PRIORITY
4 : ENABLE-HISTORY
5 : ENABLE-AUTO-HISTORY
6 : KEEP-ALIVE
7 : RESET RTU STATISTICS
8 : STOP RTU
9 : STOP LIVE
10 : DISABLE-PRIORITY
11 : DISABLE-HISTORY
12 : DISABLE-AUTO-HISTORY
13: DISABLE-KEEP-ALIVE
14 :
15 : RESET-OVERFLOW-FLAGS
This word is fleeting bits, i.e. one shots under normal conditions the action word will be 0000 indicating no change in current state.

Statistics Words:

0 : # of reboots
1 : # of retries
2 : # messages sent
3 : # messages recieved
4 : # open errors
5 : # send errors
6 : # recieve errors
7 : Last months bytes tx (internal)
8 : Last months bytes rx (internal)
9 : current months bytes tx (internal)
10 : current months bytes rx (internal)
11 : Today bytes tx (internal)
12 :Today bytes rx (internal)

StatusWord 0: *(used with all responses)*

0 : STARTED
1 : NEEDTIME { 0(no) , 1(yes) }
2 : LIVE { 0(no) , 1(yes) }
3 : PRIORITY-STARTED
4 : PRIORITY-OVER { 0(no) , 1(yes) } (should be alarmed)
5: HIST STARTED
6 : HIST-OVER { 0(no) , 1(yes) } (should be alarmed)
7 : AUTO_REPORT_HISTORY { 0(no) , 1(yes) }
8 : WILL SEND KEEPALIVES
9 : Contact time exceeded (internal)
10 : Device or Driver in minimal mode (internal)
11 : RTU vs SCADA mapping mismatch.
12,13 : 2bit*25 Count of HISTORY BUFFER
14-15 : 2bit*25 Count of UNSOLICITED BUFFER

ActionWord 1: *{ internal }*

0 : (internal) INTERROGATE NOW
1 : (internal) READ CONFIG NOW
2 : (internal) WRITE TIME NOW
3 : (internal READ HISTORY NOW
4 : (internal) READ STATISTICS NOW
5 : -
6 : -
7 : -
8 : (internal ENABLE MINIMAL device only)
9 : (internal ENABLE MINIMAL whole driver)
10: (internal DISABLE MINIMAL device only)
11: (internal DISABLE MINIMAL whole driver

These cause other functions to trigger and is not part of action word0 which is actually written to the plc

Config Area: *(Words)*

0 : ID (immediate)
1 : IP Msw (readonly - reset required)
2 : IP Lsw (readonly - reset required)
3 : PORT (readonly - reset required)
4 : TIMEOUT (immediate 100ms) [>=20]
5 : RETRIES (immediate)
6 : LIVE PERIOD (immediate 100ms)
7 : LIVE REPORT INTERVAL (immediate 100ms) [>=100]
8 : AUTO HIST PERIOD (immediate hours)
9 : AUTO HIST REPORT INTERVAL (immediate 100ms)[>=100]
10 : RTC UPDATE INTERVAL (immediate hours)
11 : KEEP ALIVE INTERVAL (immediate 100ms) [>=100]
12 : RTU SOFTWARE VERSION (readonly)
13 : NEW IP MSW (AW0 reset required)
14 : NEW IP LSW (AW0 reset required)
15 : NEW Port (AW0 reset required)

Once these are writen a reset RTU must get a RESET onACTIONW0 to initiate.
Last known good config is updated on reception of a valid master message. If a reconfig fails, the next hard reset will use last known.

These are transacted via READCFG,WRITECFG.

```
Typedef struct TIMEOBJECT
{
lsb
    lsw     unsigned Sec : 6;   //0 -> 59
            unsigned Min : 6;   //0 -> 59
            unsigned Hour : 5; //0 -> 23

            unsigned Day : 5;  //1 -> 31
            unsigned Month : 4; //1 -> 12
    msw     unsigned Year : 6; //0 -> 64 since 2000
msb
}
```

TO msw	TO lsw

FunctionWord: *(used with all transactions)*
000x --- > FUNCTION NIBBLE …. { Action requires such as interrogate }
00x0 --- > DIRECTION (0=initiated from master, 1=initiated from slave(unsolicited, or link so far))
01xx <---- ACK …. acknowledge reception.
10xx <---- NACK .... nacknowledge reception.

Function nibble 000x , Direction nibble 00x0 , Ack nibble 0x00 , Nak nibble x000

0000 ----> LINK/HELLO [ACTION WORD from master]
0010 <--- LINK/HELLO [STATUS WORD from slave]

Links don't have acks except the masters link results in a slave link reply of StatusWord of the actions taken.
Master acknowledges slaves link, as it is considered a keep alive … and the slave needs it for link through confirmation especially important on modem(gprs) links

0001 ----> INTERROGATE …. synchronous poll request (always master initiated)
0111 <---- ACK [CTO+FIXED ALL DATA] …. Ack with 1 single instantaneous data packet containing all data preceeded by a common time object to be applied to all updates to the scada.

0012 <---- UNSOLICITED [TIMEOBJ+INDEX+DATA*(number of events) {*2 if LONG}]
0102 ----> ACK

0003 ----> HISTORY … Request total emptying of history buffer via async+ack (0013 messages)
0113 <---- ACK or HISTORY [TIMEOBJ+INDEX+DATA*(number of events) {*2 if LONG}]
0103 ----> ACK
1013 <---- NACK [NO MORE DATA] this will be sent by the plc to indicate the buffer is now empty

0004 --- > WRITEDATA [OFFSET+DATA] …. Write to a single logical (RTU) item.
0114 <---- ACK …. acknowledge reception.

0005 --- > WRITECFG [OFFSET+DATA] …. Write to a single system config (RTU) item.
0115 <---- ACK …. acknowledge reception.

0006 --- > READCFG …. Fetches all cfg words in one fixed single update packet.
0116 <---- ACK [FIXED ALL DATA] …. acknowledge reception.

0007 --- > WRITETIME [FIXED TIMEOBJ + ACTIONWORD] …. Write timeobj (2 words) time to RTU (fixed)
0117 <---- ACK …. acknowledge reception.

0008 --- > READSTATS …. Fetches all plcstats words in one fixed single update packet.
0118 <---- ACK [FIXED ALL DATA] …. acknowledge reception.

1xxx <--> NAK [FIXED ERRORW]

“AD”	**“RO”**	**< packet >**

0 PREFIX AND POSTFIX

0.1 Prefix

Prefix is a 2 word header sent with every packet master and standby

“AD”	**“RO”**

0.1 Postfix

There is no postfix at this time.

1 HELLO&MASTER ACTION WRITE

0000 ----> LINK/HELLO [ACTION WORD from master]
0010 <--- LINK/HELLO [STATUS WORD from slave]

Links don't have acks except the masters link results in a slave link reply of StatusWord of the actions taken.

Master acknowledges slaves unsolicited link, as it is considered a keep alive … and the slave needs it for link through confirmation especially important on modem(gprs) links

1.1a Link/Hello message from master: *(action word is of interest or NULL then its just a keep-alive)*

Prefix +	**Station ID msb**	**Master ID lsb**	**Function (0000)**	**Action Word**

1.2a Link/Hello response message from slave: *(status word is of interest) Link can also function as keepalive*

Prefix +	**Station ID msb**	**Station ID lsb**	**Function (0110)**	**Status Word**

Note the difference 01xx Ack nibble where the later is not an ack as the plc originates the sequence.

OR

1.1b Link/Hello unsolicted message from slave: *(status word is of interest) Link can also function as keepalive*

Prefix +	**Station ID msb**	**Station ID lsb**	**Function (0010)**	**Status Word**

1.2b Ack message from master:

Prefix +	**Station ID msb**	**Station ID lsb**	**Function (0100)**

2 INTERROGATE

0001 ----> INTERROGATE …. synchronous poll request (always master initiated)
0111 <---- ACK [CTO+FIXED ALL DATA] …. Ack with 1 single instantaneous data packet containing all data preceeded by a common time object to be applied to all updates to the scada.

2.1 Data (INTERROGATE) request from master:

Prefix +	**Station ID msb**	**Station ID lsb**	**Function (0001)**

2.2a Data (INTERROGATE) message from slave:

CTO is not repeated, is applied to all indexes in this transmission.

If indexes are duplicated the type is a DD type, with
DWn = msw
DW(n+1) = lsw.

Prefix +	**Station ID msb**	**Station ID lsb**	**Function (0111)**	**Status Word**	**Count (Words)**	**CTOmsw**	**CTOlsw**	**Index0**	**DW0**	**Index1**	**DW1**

Repeated 2 or 4 words

2.2b Data (INTERROGATE) NAK message from slave:

Prefix +	**Station ID msb**	**Station ID lsb**	**Function (1011)**	**Status Word**	**ErrorW**

3 Unsolicited PRIORITY report to Master

0012 <---- PRIORITY [TIMEOBJ+INDEX+DATA*(number of events) {*2 if LONG}]
0102 ----> ACK

3.1 Data (PRIORITY) report from slave:

TO Timeobject is only applicable for the index which it prefixes.

Single word type : DWx

Prefix +

Station ID msb	Station ID lsb	Function (0012)	Status Word	Count (Words)	TOmsw	TOlsw	Index0	DW0

TO is repeated every 4 words.

TO & index is always zero for the lsw of a double type, don't confuse with interrogate where index is repeated for both the lsw and msw of double types.

Prefix +

Station ID msb	Station ID lsb	Function (0012)	Status Word	Count (Words)	TOmsw	TOlsw	Index0	DW0 msw	----	----	----	DW1 lsw

TO is repeated every 8 words.

Double word type : DDx

Single and double types can both be reported in the same packet.

3.2a ACK from master:

Prefix +

Station ID msb	Station ID lsb	Function (0102)

3.2b NACK from master:

The master will never need to NAK a properly formatted message, and thus is silent NAK, i.e. the slave will repeat the last message until a master ACK is received.

```
Typedef struct TIMEOBJECT
{
lsb
   lsw     unsigned Sec : 6;   //0 -> 59
           unsigned Min : 6;   //0 -> 59
           unsigned Hour : 5; //0 -> 23
           unsigned Day : 5;  //1 -> 31
           unsigned Month : 4; //1 -> 12
   msw     unsigned Year : 6; //0 -> 64 since 2000
msb
}
```

---

4 Solicited & Unsolicited HISTORY report to Master

0003 ----> HISTORY … Request total emptying of history buffer via unsolicited asynchronous+ack messages (0113)

0113 <---- ACK or HISTORY [TIMEOBJ+INDEX+DATA*(number of events) {*2 if LONG}]
0103 ----> ACK
1013 <---- NACK [NO MORE DATA] this will be sent by the plc to indicate the buffer is now empty

The Master can request a HISTORY dump from the slave device, which will result in a a flurry of HISTORY reports that will require an ACK, once the history buffer is empty a NAK + NOMOREDATA will be sent to the master.

The response to a master HISTORY request can be considered an ACK, i.e. a slave ACK to master will always contain history data.

Setting the PLC into LIVE mode and or AUTO-HISTORY mode will also result in unsolicited HISTORY updates being sent to the master on a periodic basis. The NAK + NOMOREDATA isstill snet after the completion of each dump session. Periodic intervals and durations are all configurable see the CW's.

4.1 Data (HISTORY) request from master:

Prefix +	Station ID msb	Station ID lsb	Function (0003)

4.2 Data (HISTORY) report from slave:
(can be unsolicted and doesn't require an initial 4.1 request) -> Function (0013)
(if in response to a 4.1 request from master) ----------------------> Function (0113)

TO Timeobject is only applicable for the index which it prefixes.

Single word type : DWx

Prefix +	Station ID msb	Station ID lsb	Function (0x13)	Status Word	Count (Words)	TOmsw	TOlsw	Index0	DW0

TO is repeated every 4 words.

TO & index is always zero for the lsw of a double type, don't confuse with interrogate where index is repeated for both the lsw and msw of double types.

Prefix +

Station ID msb	Station ID lsb	Function (0x13)	Status Word	Count (Words)	TOmsw	TOlsw	Index0	DW0 msw	----	----	----	DW1 lsw

TO is repeated every 8 words.

Double word type : DDx

Single and double types can both be reported in the same packet.

4.3a ACK from master:

Prefix +	Station ID msb	Station ID lsb	Function (0103)

4.3b NACK from master:

The master will never need to NAK a properly formatted message, and thus is silent NAK, i.e. the slave will repeat the last message until a master ACK is received.

```
Typedef struct TIMEOBJECT
{
lsb
    lsw     unsigned Sec : 6;   //0 -> 59
            unsigned Min : 6;   //0 -> 59
            unsigned Hour : 5; //0 -> 23
            unsigned Day : 5;  //1 -> 31
            unsigned Month : 4; //1 -> 12
    msw     unsigned Year : 6; //0 -> 64 since 2000
msb
}
```

4.4 NACK + NOMOREDATA from slave:

Station ID msb	Station ID lsb	Function (1013)	Status Word	ErrorW (0005)

Once the PLC's history buffer is empty (or exceeded maximum packets per session) a 4.4 Nack + nomoredata will be sent.

5 WRITEDATA

0004 --- > WRITEDATA [OFFSET+DATA] …. Write to a single (DW) or double (DD) (RTU) item.
0114 <---- ACK …. acknowledge reception.

5.1 WRITEDATA control from master:

Count is in words

Prefix +	Station ID msb	Station ID lsb	Function (0004)	Offset	Count (0001)	DW

Prefix +	Station ID msb	Station ID lsb	Function (0004)	Offset	Count (0002)	DD lsw	DD msw

NB – lsw of the DD is written first,

5.2 ACK from slave:

Prefix +	Station ID msb	Station ID lsb	Function (0114)	Status Word

Controls are retried without user intervetion According to the number of retries configured.

5.3 NACK from slave:

Prefix +	Station ID msb	Station ID lsb	Function (1014)	Status Word	ErrorW

---

6 WRITECONFIG

0005 --- > WRITECFG [OFFSET+DATA] …. Write to a single system config (DW) (RTU) item.
0115 <---- ACK …. acknowledge reception.

There is **no** DD write command for Configuration data area
The config buffer is a mixed DW and DD buffer, but since all immediate indexes are DW only therefore the write takes effect immediately.
The DD indexs that require 2 words are writen seperately and need to be activated via a reset, make sure the application knows to write both words before initiating a reset.

6.1 WRITECONFIG control from master:

Count is in words

Prefix +	Station ID msb	Station ID lsb	Function (0005)	Offset	Count (0001)	DW

6.2 ACK from slave:

Prefix +	Station ID msb	Station ID lsb	Function (0115)	Status Word

Controls are retried without user intervetion According to the number of retries configured.

6.3 NACK from slave:

Prefix +	Station ID msb	Station ID lsb	Function (1015)	Status Word	ErrorW (0005)

---

7 WRITETIME

0007 --- > WRITETIME [FIXED TIMEOBJ + ACTIONWORD] …. Write timeobj (2 words) time to RTU (fixed)
0117 <---- ACK …. acknowledge reception.

See the TimeStruct for the bit packed definition of the timestructure.
This is the same timestructure used on timebased objects (TO)

Controls are retried without user intervetion According to the number of retries configured.

7.1 WRITETIME control from master:

Not used at this time forced to (0000)

Prefix +	Station ID msb	Station ID lsb	Function (0007)	TimeObj Msw	TimeObj Lsw	Action Word

Typedef struct **TIMEOBJECT**
{
lsb
lsw unsigned Sec : 6; //0 -> 59
unsigned Min : 6; //0 -> 59
unsigned Hour : 5; //0 -> 23
unsigned Day : 5; //1 -> 31
unsigned Month : 4; //1 -> 12
msw unsigned Year : 6; //0 -> 64 since 2000
msb
}

7.2 ACK from slave:

Prefix +	Station ID msb	Station ID lsb	Function (0115)	Status Word

7.3 NACK from slave:

Prefix +	Station ID msb	Station ID lsb	Function (1015)	Status Word	ErrorW (0005)

---

8 READCONFIG

0006 --- > READCFG …. Fetches all cfg words in one fixed single update packet.
0116 <---- ACK [FIXED ALL DATA] …. acknowledge reception.

8.1 READCONFIG Request from master:

Prefix +	Station ID msb	Station ID lsb	Function (0006)

8.2 READCONFIG response from slave:

Prefix +	Station ID msb	Station ID lsb	Function (0116)	Status Word	Count (000F)	DW0	…	DW15

Fixed 16 word buffer

8.3 READCONFIG NAK message from slave:

Prefix +	Station ID msb	Station ID lsb	Function (1016)	Status Word	ErrorW

---

9 READSTATS

0008 --- > READSTATS …. Fetches all plcstats words in one fixed single update packet.
0118 <---- ACK [FIXED ALL DATA] …. acknowledge reception.

9.1 READSTATS Request from master:

Prefix +	Station ID msb	Station ID lsb	Function (0008)

9.2 READSTATS response from slave:

Prefix +	Station ID msb	Station ID lsb	Function (0118)	Status Word	Count (0007)	DW0	…	DW6

9.3 READSTATS NAK message from slave:

Prefix +	Station ID msb	Station ID lsb	Function (1018)	Status Word	ErrorW

Statistics Words:

0 : # of reboots
1 : # of retries
2 : # messages sent
3 : # messages recieved
4 : # open errors
5 : # send errors
6 : # recieve errors
7 : Last months bytes tx (internal)
8 : Last months bytes rx (internal)
9 : current months bytes tx (internal)
10 : current months bytes rx (internal)
11 : Today bytes tx (internal)
12 :Today bytes rx (internal)

Fixed 0-6 word buffer, where only the first 7 words are sent from the PLC.
(Word 7 through 12) are driver internal and not directly returned on the protocol

---

If any protocol message is not implemented, a NAK [huh?] Will be at least expected.